Chapter 1

Discrete Mathematics

1.1 Objectives

Content adapted from [Piff(1991), Rosen(2007), Giunchiglia and Fumagalli(2017b)]. Please read first five chapters of [Piff(1991)] and first two chapters of [Rosen(2007)].

1. Defining Propositional Logic.
2. Defining Predicate Logic.
3. Defining Set, function, or relation model, and interpret the associated operations.
4. Operations between sets, functions, and relations.
5. Constructing proofs.
6. Logical reasoning to solve a strategic problem.
7. Model a real-life industry application applying formal methods.

1.2 Proposition Logic

Definition 1 (Proposition)**.** *A proposition is a statement that is either true or false.*

1.2.1 Examples of propositions

1. Blue is a color.
2. $1 + 1 = 2$.
3. Any two primes have a common factor.

1.2.2 Examples of sentences that are not propositions

1. What is the color of sky?

2. $1 + 1$.

3. This sentence is false.

4. $x^2 = x$.

5. There exists an x such that, $x^2 = x$.

6. For all integers x, $x^2 = x$.

Definition 2 (Propositional variables)**.** *A proposition or a boolean variable is a variable that is either true or false.*

Example 1. *The following symbolic sentence is read as x squared is equal to y.*

$$P(x, y) := x^2 = y$$

. This is not a proposition.

However, for a given x and y, the sentence becomes a proposition.

$$P(x = 1, y = 1) := 1^2 = 1$$

1.3 Propositional calculus aka Boolean Algebra

In propositional calculus, we have different names for Boolean algebra operators.

Boolean algebra	Propositional calculus
OR gate $x + y$	Disjunction: $P \vee Q$
AND gate $x \cdot y$	Conjunction: $P \wedge Q$
NOT gate $\bar{y}$	Negation: $\neg P$

1.3.1 Implication $\implies$, $\supset$, $\rightarrow$

$$(P \rightarrow Q) \equiv ((\neg P) \vee Q)$$

1.3.2 Equivalence $\equiv$, $\leftrightarrow$

$$P \equiv Q =$$

1.4 Propositional logic language

Definition 3. *Propositional alphabet*

Logic symbols $\neg, \wedge, \vee, \supset, \equiv$

Propositional variables $p \in \mathcal{P}$

Separator symbols *"(" and ")"*

Definition 4. *Well formed formulas*

- *every* $p \in \mathcal{P}$ *is an atomic formula*
- *if* A *and* B *are formulas then* $\neg A, A \wedge B, A \vee B, A \supset B$ *and* $A \equiv B$ *are formulas.*

Example 2. *Lets consider a propositional language where p means Paola is happy, q means Paola paints a picture, and r means Renzo is happy. Formalize the following sentences:*

1. *if Paola is happy and paints a picture then Renzo isnt happy*
2. *if Paola is happy, then she paints a picture*
3. *Paola is happy only if she paints a picture*

The precision of formal languages avoid the ambiguities of natural languages.

Definition 5. *A Propositional interpretation is a function* $I : P \rightarrow \{True, False\}$. *(Evaluation of the formula with each variable set to true or false.)*

Definition 6. *A formula A is*
Valid if for all interpretations I , $I \models A$
Satisfiable if there is an interpretations I *s.t.,* $I \models A$
Unsatisfiable if for no interpretations I , $I \models A$

Propositional logic is an instrument for writing proofs. There are two connectives between two consecutive statements.

1.4.1 Equivalence $\iff$

Example 3 ([Giunchiglia and Fumagalli(2017a)])**.** *Socrate says:*
If Im guilty, I must be punished; I must not be punished. Thus Im not guilty.
Is the argument logically correct?

Example 4 ([Giunchiglia and Fumagalli(2017a)]). *Socrate says:*
If Im guilty, I must be punished; I must be punished. Thus Im guilty. Is the argument logically correct?

Example 5. *As a reward for saving his daughter from pirates, the King has given you the opportunity to win a treasure hidden inside one of three trunks. The two trunks that do not hold the treasure are empty. To win, you must select the correct trunk. Trunks 1 and 2 are each inscribed with the message This trunk is empty, and Trunk 3 is inscribed with the message The treasure is in Trunk 2. The Queen, who never lies, tells you that only one of these inscriptions is true, while the other two are wrong. Which trunk should you select to win?*

Problem 1. *(20 marks) Suppose that in Example 7, the inscriptions on Trunks 1, 2, and 3 are The treasure is in Trunk 3, The treasure is in Trunk 1, and This trunk is empty. For each of these statements, determine whether the Queen who never lies could state this, and if so, which trunk the treasure is in.*
a) All the inscriptions are false.
b) Exactly one of the inscriptions is true.
c) Exactly two of the inscriptions are true.
d) All three inscriptions are true.

Problem 2 ([Giunchiglia and Fumagalli(2017a)]). *(20 marks) Kyle, Neal, and Grant find themselves trapped in a dark and cold dungeon (HOW they arrived there is another story). After a quick search the boys find three doors, the first one red, the second one blue, and the third one green. Behind one of the doors is a path to freedom. Behind the other two doors, however, is an evil fire-breathing dragon. Opening a door to the dragon means almost certain death.*

On each door there is an inscription:

freedom is behind this door	*freedom is not behind this door*	*freedom is not behind the blue door*

Given the fact that at LEAST ONE of the three statements on the three doors is true and at LEAST ONE of them is false, which door would lead the boys to safety?

1.5 Predicates

Definition 7 (Predicates). *A predicate is a statement* $P(x_1, \dots, x_n)$ *involving* n *(not necessarily boolean) variables* $x_1 \in \mathcal{X}_1$*,* $x_n \in \mathcal{X}_n$*, and which evaluates to a proposition* $P(x_1, \dots, x_n) \in \{True, False\}$*.*

Example 6. *Let P(x) denote the statement x > 3. What are the truth values of P(4) and P(2)?*

1.5.1 Quantifiers

Definition 8 (Universal quantifier $\forall$).

Definition 9 (Existential quantifier $\exists$).

Example 7. *Let Q(x) be the statement x < 2. What is the truth value of the quantification* $\forall x Q(x)$*, where the domain consists of all real numbers?*

Example 8. *Let Q(x) denote the statement x = x + 1. What is the truth value of the quantification* $\exists x Q(x)$*, where the domain consists of all real numbers?*

1.5.2 Rules of Inference for Propositional Logic

TABLE 1 Rules of Inference.

Rule of Inference	*Tautology*	*Name*
p $\underline{p \to q}$ $\therefore q$	$(p \land (p \to q)) \to q$	Modus ponens
$\neg q$ $\underline{p \to q}$ $\therefore \neg p$	$(\neg q \land (p \to q)) \to \neg p$	Modus tollens
$p \to q$ $\underline{q \to r}$ $\therefore p \to r$	$((p \to q) \land (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \lor q$ $\underline{\neg p}$ $\therefore q$	$((p \lor q) \land \neg p) \to q$	Disjunctive syllogism
$\underline{p}$ $\therefore p \lor q$	$p \to (p \lor q)$	Addition
$\underline{p \land q}$ $\therefore p$	$(p \land q) \to p$	Simplification
p $\underline{q}$ $\therefore p \land q$	$((p) \land (q)) \to (p \land q)$	Conjunction
$p \lor q$ $\underline{\neg p \lor r}$ $\therefore q \lor r$	$((p \lor q) \land (\neg p \lor r)) \to (q \lor r)$	Resolution

Example 9. *Give a direct proof of the theorem If n is an odd integer, then n^2 is odd.*

1.6 Sets

The concept of *set* is considered a primitive concept in math

A set is a collection of *elements* whose description must be unambiguous and unique: it must be possible to decide whether an element belongs to the set or not.

1.6.1 Describing sets

Listing

Abstraction

Eulero-Venn Diagrams

1.6.2 Russell's paradox

$$R = \{x \mid x \notin x\}. Then R \in R \iff R \notin R.$$

1.6.3 Relationships between sets

Definition 10 (Empty Set $\emptyset$)**.** $\emptyset = \{\}$

Definition 11 (Singleton $\{a\}$)**.** $A = \{a\}$ *is singleton and* $\{a\} \neq a$.

Definition 12 (Set membership $a \in A$)**.** $A = \{a, b, c\} \implies a \in A$ *and* $d \notin A$

Definition 13 (Set equality $A = B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a\} \implies A = B$.

Definition 14 (Subset $A \subseteq B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a\} \implies A \subseteq B$.
$A = \{a, b, c\}$ $B = \{c, b, a, a, d\} \implies A \subseteq B$.

Definition 15 (Subset $A \subset B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a, d\} \implies A \subset B$.
$A = \{a, b, c\}$ $B = \{c, b, a, a\} \implies A \not\subset B$.

Definition 16 (Power set $P(A)$, 2^A)**.** $A = \{a, b, c\}$, *then* $P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}$.

Definition 17 (Union of two sets $A \cup B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a, d\}$ $\implies A \cup B = \{a, b, c, d\}$.

Definition 18 (Intersection of two sets $A \cap B$)**.** $A = \{a, b, c\}$ $B = \{c, b, a, a, d\}$ $\implies A \cap B = \{a, b, c\}$.

Definition 19 (Set difference $A \setminus B$, $A - B$). $A = \{a, b, c\}$ $B = \{c, b, a, a, d\}$ $\implies A \setminus B = A - B = \{\} = \emptyset$.
$B \setminus A = B - A = \{d\}$.

Definition 20 (Set complement $\bar{A}$, $C_U A$). *Given a universal set U, complement is defined as $\bar{A} = C_U A = U - A$. Also, $A \cup \bar{A} = U$.*

Example 10. *Prove that $U = (A \cap B) \cup \overline{(A \cap B)}$*

Example 11. $A = \{a, e, i, o, \{u\}\}$ *and* $B = \{i, o, u\}$

1. $B \in A$
2. $(B - \{i, o\}) \in A$
3. $\{a\} \cup \{i\} \subset A$
4. $\{u\} \in A$
5. $\{\{u\}\} \in A$
6. $B - A = \emptyset$
7. $i \in A \cap B$
8. $\{i, o\} = A \cap B$

Example 12. *Prove De Morgan laws for sets $\overline{A \cap B} = \bar{A} \cup \bar{B}$, and $\overline{A \cup B} = \bar{A} \cap \bar{B}$.*

1.7 Relations

Definition 21 (Cartesian product of sets $A \times B$). $A \times B = \{(a, b) : \forall a \in A, \forall b \in B\}$

Definition 22 (Relation $R \subseteq A \times B$). *We write aRy to read as "a is R-related to b".*

Example 13. *given* $A = \{3, 5, 7\}$*,* $B = 2, 4, 6, 8, 10, 12$ *and* aRb *iff* a *is a divisor of* b*, then* $R = \{(3, 6), (3, 12), (5, 10)\}$

Definition 23 (Domain of R)**.**

$$dom(R) = \{a \in A : \exists y \in B(aRy)\}$$

Definition 24 (Image/Range of R)**.**

$$Range(R) = Im(R) = \{b \in B : \exists x \in A(xRb)\}$$

Definition 25 (Inverse of R)**.**

$$R^{-1} = \{(b, a)|(aRb)\}$$

1.8 Functions

Definition 26 (Function $f : A \to B$)**.** *Given two sets A and B, a function f from A to B is a relation that associates to each element a in A exactly one element b in B. Denoted with*
$f : A \to B$

1.8.1 Classes of Functions

Definition 27 (Surjective)**.**

Definition 28 (Injective)**.**

Definition 29 (Bijective)**.** *Both Surjective and Bijective*

Definition 30 (Inverse)**.**

Definition 31 (Composition)**.** $f : A \to B$*,* $g : B \to C$*.* $g \circ f : A \to C$*;* $(g \circ f)(a) = g(f(a))$

1.9 More exercises

Example 14. *Define an appropriate language and formalize the following sentences using First Order Logic formulas.*

1. *Bill has at least one sister.*
2. *Bill has no sister.*
3. *Bill has at most one sister.*
4. *Bill has (exactly) one sister.*
5. *Bill has at least two sisters.*
6. *Every student takes at least one course.*
7. *Only one student failed Geometry.*
8. *No student failed Geometry but at least one student failed Analysis.*
9. *Every student who takes Analysis also takes Geometry.*

Example 15. *Consider the following sentences:*

1. *All actors and journalists invited to the party are late.*
2. *There is at least a person who is on time.*
3. *There is at least an invited person who is neither a journalist nor an actor.*

Formalize the sentences and prove that 3. is not a logical consequence of 1 . and 2.

Example 16. *Let $\{c_1, .., c_k\}$ be a non empty and finite set of colors. A partially colored directed graph is a structure $\langle N, R, C\rangle$ where*

- *N is a non empty set of nodes*
- *R is a binary relation on N*
- *C associates colors to nodes (not all the nodes are necessarily colored,. and each node has at most one color)*

Provide a first order language and a set of axioms that formalize partially colored graphs. Show that every model of this theory correspond to a partially colored graph, and vice-versa. For each of the following properties, write a formula which is true in all and only the graphs that satisfies the property:

1. *connected nodes don't have the same color*
2. *the graph contains only 2 yellow nodes*
3. *starting from a red node one can reach in at most 4 steps a green node*
4. *for each color there is at least a node with this color*
5. *the graph is composed of $|C|$ disjoint non empty subgraphs, one for each color*

Problem 3. *(40 marks) Minesweeper is a single-player computer game invented by Robert Donner in 1989. The object of the game is to clear a minefield without detonating a mine. The game screen consists of a rectangular field of squares. Each square can be cleared, or uncovered, by clicking on it. If a square that contains a mine is clicked, the game is over. If the square does not contain a mine, one of two things can happen: (1) A number between 1 and 8 appears indicating the amount of adjacent (including diagonally-adjacent) squares containing mines, or (2) no number appears; in which case there are no mines in the adjacent cells. An example of game situation is provided in the following figure:*
Provide a first order language that allows to formalize the knowledge of a player in a game state. In such a language you should be able to formalize the following knowledge:

1. *there are exactly* n *mines in the minefield*
2. *if a cell contains the number 1 , then there is exactly one mine in the adjacent cells.*

3. *show by means of deduction that there must be a mine in the position* $(3, 3)$ *of the game state of picture 3.1*

Suggestion: define the predicate Adj(x, y *) to formalize the fact that two cells* x *and* y *are adjacent*

Problem 4. *(40 marks)Formalize in first order logic the train connections in Italy. Provide a language that allows to express the fact that a town is directly connected (no intermediate train stops) with another town, by a type of train (e.g., intercity, regional, interregional). Formalize the following facts by means of axioms:*

1. *There is no direct connection from Rome to Trento*
2. *There is an intercity from Rome to Trento that stops in Firenze, Bologna and Verona.*
3. *Regional trains connect towns in the same region*
4. *Intercity trains don't stops in small towns.*

Bibliography

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